

Comparative Analysis of Lion and AdamW Optimizers for Cross-Encoder Reranking with MiniLM, GTE, and ModernBERT

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Introduction

Research Overview & Motivation

- **Information Retrieval Challenge:** Modern IR systems require precise document ranking from vast collections
- **Cross-Encoder^[2] Advantage:** State-of-the-art reranking through deep query-document interactions
- **Optimization Gap:** Limited research on Lion optimizer^[4] for IR tasks despite success in other domains
- **Practical Impact:** Optimizer choice significantly affects training cost and model performance

Research Problem:

How do Lion^[4] and AdamW^[3] optimizers compare across different cross-encoder architectures for information retrieval tasks?

Research Objectives

- 1. Systematic Evaluation:** Compare Lion^[4] and AdamW^[3] across three transformer architectures.
- 2. Comprehensive Benchmarking:** Evaluate performance using standard IR metrics on TREC DL 2019^[8] and MS MARCO^[9].
- 3. Training Dynamics Analysis:** Investigate convergence patterns and stability.
- 4. Practical Guidelines:** Provide evidence-based recommendations for optimizer selection.

Expected Contributions

- Empirical evidence comparing optimizers across multiple architectures
- Optimization guidelines for cross-encoder training in IR
- Analysis of training efficiency and resource utilization

Literature Review

BERT & Cross-Encoder Architecture

BERT-based Cross-Encoders^[2] for Information Retrieval

Key Innovation:

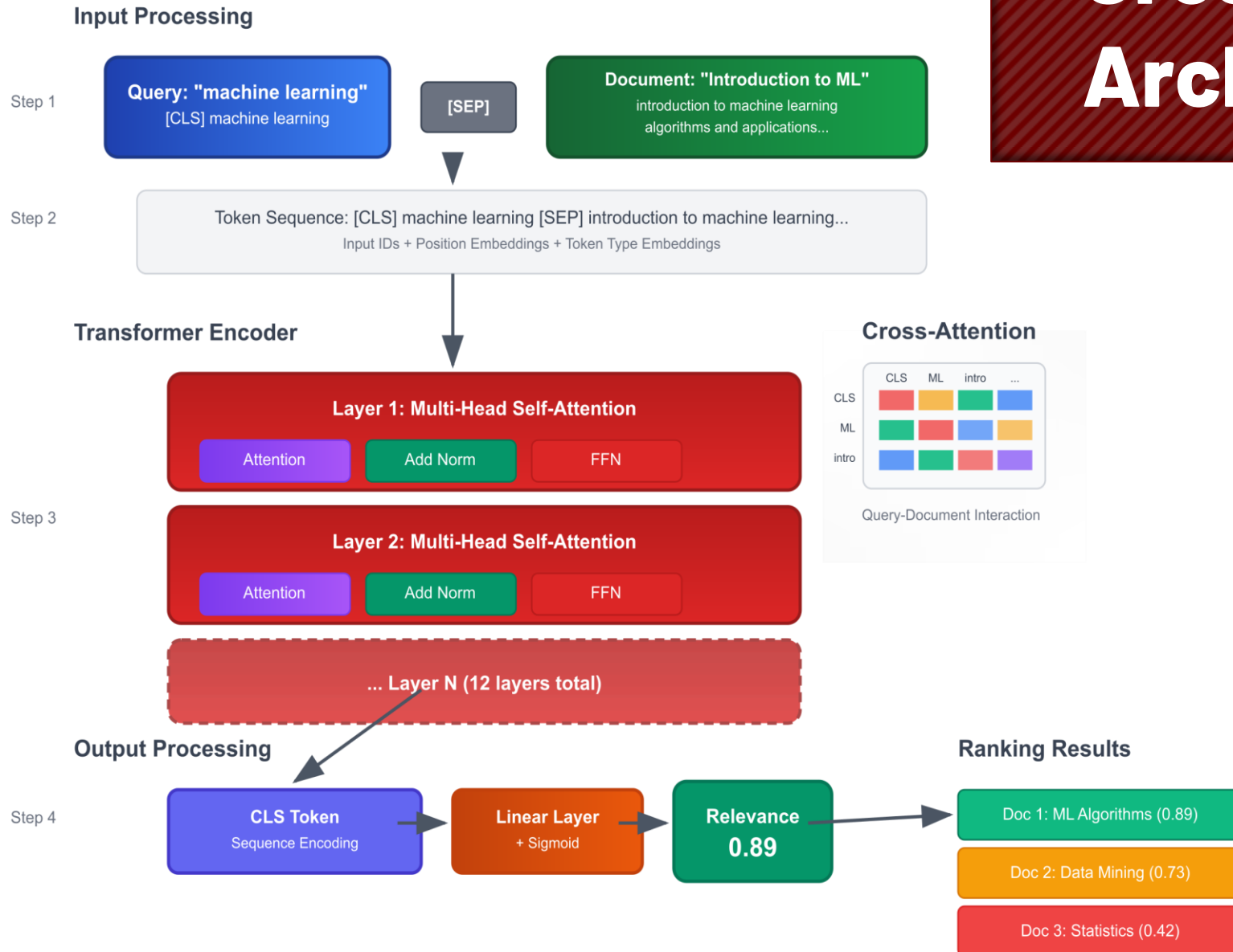
- Joint encoding: [CLS] Query [SEP] Document [SEP]
- Fine-tuning for relevance prediction
- Superior to bi-encoders for reranking

Modern Variants:

- MiniLM^[5]: Distilled BERT for efficiency
- GTE^[7]: Multilingual, extended context (8192 tokens)
- ModernBERT^[6]: Advanced features (RoPE^[12], Flash Attention^[13])

Standard approach for document reranking tasks

Cross-Encoder Architecture



Key Advantages

- ❑ Deep Interaction
- ❑ Token-level Attention
- ❑ Superior Accuracy

AdamW: The Standard Optimizer ^[3]

Evolution: Adam $\rightarrow$ AdamW

Key Improvement:

- Decoupled weight decay from gradient updates

Why AdamW Dominates:

- ✓ Better generalization than Adam
- ✓ Stable training for transformers
- ✓ De facto standard for BERT^[1] family models

Current Status: Default choice without alternatives explored

AdamW Algorithm

AdamW Optimizer

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t \quad (1)$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 \quad (2)$$

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t} \quad (3)$$

$$\hat{v}_t = \frac{v_t}{1 - \beta_2^t} \quad (4)$$

$$\theta_t = \theta_{t-1} - \eta \left(\frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} + \lambda \theta_{t-1} \right) \quad (5)$$

where:

g_t : gradient at time step t

m_t : exponential moving average of gradients (momentum)

v_t : exponential moving average of squared gradients

η : learning rate

λ : weight decay coefficient

β_1, β_2 : exponential decay rates for moment estimates

Lion: A Novel Alternative^[4]

Revolutionary Approach:

- Sign-based momentum
- Simplified computation vs Adam family
- Reduced memory requirements

Proven Results:

- ✓ Competitive performance in computer vision
- ✓ Lower computational overhead
- ✓ Faster convergence in some tasks

Research Gap:

- ✗ Unexplored in NLP/Information Retrieval
- ✗ No systematic comparison with AdamW for cross-encoders

Lion Algorithm

Lion Optimizer

$$c_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t \quad (1)$$

$$\theta_t = \theta_{t-1} - \eta (\text{sign}(c_t) + \lambda \theta_{t-1}) \quad (2)$$

$$m_t = \beta_2 m_{t-1} + (1 - \beta_2) g_t \quad (3)$$

where:

c_t : interpolated momentum for update direction

$\text{sign}(\cdot)$: element-wise sign function

θ_t : model parameters at time step t

g_t : gradient at time step t

m_t : exponential moving average of gradients (momentum)

η : learning rate

β_1 : momentum coefficient for update direction (typically 0.9)

β_2 : momentum coefficient for gradient tracking (typically 0.99)

λ : weight decay coefficient

Methodology

Experimental Setup

Dataset Configuration

Training: MS MARCO^[9] Passage Ranking (~500K query-passage pairs)

Evaluation: TREC DL^[8] 2019 (43 queries with comprehensive relevance judgments)

Metrics: NDCG@10, MAP, MRR@10, Recall@10, R-Precision, P@10

Infrastructure

Platform: Modal cloud computing^[10]

Hardware: 3 × NVIDIA L40S GPUs

Monitoring: Weights & Biases^[11]

Training: 3 epochs per configuration

Experimental Design

Total Configurations: 6 (3 models × 2 optimizers)

Controlled Variables: Batch size, training steps, evaluation frequency

Variable Parameters: Learning rates (2e-5 standard, 2e-6 for ModernBERT+Lion)



Model Architectures Tested

Model	# Parameters	Context Length	Key Features
MiniLM-L12-H384 ^[5]	~33M	512 tokens	Efficient baseline, knowledge distillation
GTE-multilingual-base ^[7]	~305M	8192 tokens	Extended context, multilingual support
ModernBERT-base ^[6]	~149M	8192 tokens	RoPE ^[12] , Flash Attention ^[13] , GeGLU ^[14]

MiniLM: Efficiency-focused for resource-constrained scenarios

GTE: Extended context for comprehensive document analysis

ModernBERT: State-of-the-art with advanced architectural innovations

Hyperparameter Settings

Parameter	AdamW	Lion
Learning Rate	2e-5 (standard)	2e-5 (2e-6 for ModernBERT ^[6])
Weight Decay	0.01	0.01
Beta1	0.9	0.9
Beta2	0.999	0.99
Scheduler	Linear	Cosine Annealing ^[15] (ModernBERT)

Evaluation Metrics

NDCG@10 - Normalized Discounted Cumulative Gain at rank 10^[16]

NDCG@10 measures the quality of ranking systems by considering both the relevance of retrieved documents and their positions in the ranking. It is normalized by the ideal DCG to ensure values between 0 and 1, where higher values indicate better performance.

$$\text{NDCG@10} = \frac{\text{DCG@10}}{\text{IDCG@10}} \quad (1)$$

where:

$$\text{DCG@10} = \sum_{i=1}^{10} \frac{2^{rel_i} - 1}{\log_2(i + 1)} \quad (2)$$

and rel_i is the relevance score of the document at position i , and IDCG@10 is the ideal DCG computed using the best possible ranking.

Evaluation Metrics

MRR@10 (Mean Reciprocal Rank at rank 10)

MRR@10 measures the average of reciprocal ranks of the first relevant document found within the top 10 results across all queries. It emphasizes finding at least one relevant document early in the ranking.

$$\text{MRR@10} = \frac{1}{|Q|} \sum_{q=1}^{|Q|} \frac{1}{\text{rank}_q} \quad (1)$$

where rank_q is the position of the first relevant document for query q within the top 10 results. If no relevant document is found in the top 10, the reciprocal rank is 0.

Results & Discussion

Main Results

Model	Optimizer	NDCG@10	MAP	MRR@10	Recall@10	P@10
MiniLM	AdamW	0.7127	0.4908	0.5826	0.1715	0.8093
	Lion	0.7031	0.4858	0.5988	0.1724	0.8116
GTE	AdamW	0.7224	0.5005	0.5942	0.1733	0.8163
	Lion	0.6909	0.4921	0.5957	0.1721	0.8140
ModernBERT	AdamW	0.7105	0.5066	0.5916	0.1678	0.8163
	Lion	🏆 0.7225	🏆 0.5121	🏆 0.5988	0.1732	🏆 0.8256



Overall Winner: ModernBERT + Lion achieved highest performance in 4/5 metrics

GPU Usage Statistics

***All calculations are w.r.t. AdamW**

Model	Optimizer	Mean Usage (%)	Peak Usage (%)	Efficiency gain (%)
MiniLM	AdamW	33.09	35.25	+2.67 %
	Lion	32.21	34.64	
GTE	AdamW	73.04	78.08	+10.33 %
	Lion	65.50	69.69	
ModernBERT	AdamW	77.04	81.40	+3.49%
	Lion	74.35	79.45	

Lion optimizer demonstrates superior GPU utilization efficiency across all models

Key Takeaways & Decision Points

Major Findings Summary

-  **Universal Efficiency:** Lion optimizer reduces GPU usage in 100% of configurations
-  **Performance Trade-offs:** Neither optimizer dominates all metrics
-  **Configuration Critical:** Learning rate and scheduling significantly impact results
-  **ModernBERT Excellence:** Best overall performance with Lion optimizer
-  **Cost-Effectiveness:** 5.50% average efficiency improvement across models



Decision Matrix for Practitioners

Priority	Model Choice	Optimizer	Expected Outcome
Maximum Performance	ModernBERT	AdamW	Best metrics
Cost Optimization	GTE	Lion	10.33% resource savings
Balanced Approach	MiniLM	Lion	Good performance + efficiency
Risk Minimization	Any	AdamW	Established, reliable results

Future Research Directions

Immediate Extensions

- Comprehensive hyperparameter studies
- Extended training analysis
- Alternative schedulers investigation
- Cross-lingual evaluation



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